The Geography of Care in Michigan: Geographic Mismatch and Policy Levers

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Abstract

Michigan faces growing concerns about nurse shortages, particularly in rural and aging communities, yet evidence on the determinants of local nurse supply remains limited. Using county-level panel healthcare workforce data from 2012–2023, we estimate two-way fixed-effects models to examine the geographic distribution of Registered Nurses (RNs) and Licensed Practical Nurses (LPNs) across Michigan's 83 counties. The strongest finding is that local LPN training output predicts subsequent LPN workforce supply, while RN training output shows no statistically significant relationship with county-level RN supply, consistent with greater geographic mobility among RNs. Educational attainment is positively associated with nurse supply; demographic demand and healthcare capacity measures show less consistent effects. A Bartik instrumental variable approach to identify wage effects was unsuccessful due to weak instrument strength. These findings suggest that expanding local training capacity may be effective for addressing LPN shortages, while improving RN availability likely requires both training and retention policies.

Contents

1 Introduction

Michigan faces a nursing workforce problem that is best understood as a geographic mismatch rather than a simple statewide shortage. National and state-level projections show that registered nurse (RN) shortages are expected to persist into the 2030s, with HRSA projecting an 18 percent RN shortage in Michigan by 2038 (National Center for Health Workforce Analysis, 2025). Yet statewide figures obscure the county-level variation most relevant for policy. The literature on healthcare workforce maldistribution shows that shortage areas are not randomly distributed: they tend to align with rurality, poverty, low educational attainment, older populations, and other markers of structural disadvantage (Streeter et al., 2020). Nursing-specific evidence similarly shows that federally designated nurse-shortage counties have historically been concentrated in nonmetropolitan areas (Stratton et al., 1995). The core question, then, is not only whether Michigan has enough nurses in aggregate, but whether RN and licensed practical nurse (LPN) supply is located where population need is greatest.

Standard policy responses often emphasize wages, loan incentives, and recruitment programs, but the labor-supply literature suggests that these tools have limited reach. Askildsen et al. (2003) estimate nurse labor-supply elasticities near 0.2, implying that moderate wage increases generate only modest increases in hours worked among employed nurses. Hanel et al. (2014) show that wage responsiveness is larger once entry into and exit from nursing are included, but this effect operates mainly by drawing nursing qualification holders back into the profession rather than by substantially expanding hours among the current workforce. At the employer level, Staiger et al. (2010) find short-run firm-level supply elasticities close to zero, consistent with monopsonistic wage-setting in local hospital labor markets. These constraints are likely most binding in rural counties, where smaller labor markets and dominant healthcare employers limit the ability of wage policy alone to correct maldistribution.

Measuring demand is also difficult because observed healthcare utilization is not the same as underlying need. In rural and underserved areas, utilization may be low because care is unavailable, unaffordable, geographically distant, or otherwise inaccessible (Reilly, 2021). Evidence from a large Midwestern healthcare system supports this concern: Nuako et al. (2022) find that rural clinic patients had worse measured health status than urban clinic patients, yet lower outpatient utilization after adjustment for demographic characteristics and health status. At the same time, the supplier-induced demand literature warns that utilization may partly reflect provider supply in high-capacity areas rather than only patient need (Evans, 1974; Grytten and Sørensen, 2001). For this reason, our county-level analysis relies on demographic and morbidity-based proxies for structural demand rather than utilization alone, a choice that motivates the data construction described in Section 4 and the conceptual framework developed in Section 3.

This paper examines the geographic distribution of RN and LPN supply across Michigan's 83 counties and estimates the county characteristics associated with differences in nurse supply. Using a county-year panel with two-way fixed effects, we relate nurse supply per 10,000 residents to structural demand measures, local training pipeline capacity, healthcare capacity, and socioeconomic disadvantage. The analysis distinguishes between counties where shortage pressure is primarily demand-driven, reflecting aging and medically complex populations, and counties where supply constraints are more closely tied to rurality, weak labor markets, and limited training infrastructure. This distinction matters for policy because expanding training capacity, improving retention, and targeting underserved counties are not interchangeable solutions. The remainder of the paper proceeds as follows. Section 2 reviews the literature on geographic shortage distribution, labor supply, demand measurement, and workforce forecasting. Section 3 develops a conceptual model of county-level nurse supply as the joint outcome of demand-side and supply-side forces. Section 4 describes the construction of the Michigan county panel. Section 5 presents the empirical framework, and Section 6 reports results. Section 7 discusses limitations, and Section 8 concludes.

2 Literature Review

The geographic concentration of healthcare workforce shortages is well established in the literature, yet most national analyses aggregate at a level that obscures the within-state variation most relevant for policy. This gap matters because the central result of this paper, that local training output predicts LPN supply but not RN supply, and the conditioning-only role assigned to capacity and shortage-designation variables in Section 6, can only be motivated by a literature that first establishes how shortages are distributed geographically and why standard policy levers operate unevenly across that geography. Streeter et al. (2020) offers a comprehensive county-level account of this maldistribution, finding that 96% of U.S. counties were either designated primary care Healthcare Professional Shortage Areas (pcHPSA) or carried at least one social disadvantage marker in 2017. Shortage status, they demonstrate, does not arise independently of structural disadvantage. The most common co-occurring markers across pcHPSA counties were rurality, low educational attainment, low income, poverty, and small population size, indicating that provider shortages cluster systematically in counties already disadvantaged along multiple dimensions. Within Health and Human Services (HHS) Region 5, which includes Michigan, poverty and an older population were the dominant sources of county-level variability, and only 4.2% of Region 5 counties exhibited more than six co-occurring disadvantage markers, a markedly lower share than Region 4 (39%) or Region 6 (36%). Michigan thus occupies a moderate position in the national distribution. Critically, however, Streeter et al. (2020) explicitly acknowledge that their cross-sectional, national design cannot capture sub-state variation and call for replication at finer geographic levels, a gap this paper addresses directly.

Nursing-specific evidence reinforces this picture while sharpening the geographic focus. Stratton et al. (1995), drawing on county-level data from the Area Health Resource File (AHRF), find that 92.4% of federally designated nurse-shortage counties in 1990 were non-metropolitan, with the largest concentration in rural counties not adjacent to any metropolitan area. This suggests that shortages reflect structural regional maldistribution as much as aggregate supply-demand imbalances. The economic and demographic disparities between rural and urban places, they argue, are directly mirrored in shortage and non-shortage counties alike. Michigan's own 2025 Healthcare Workforce Index underscores the contemporary severity of these pressures: 35.5% of Michigan RNs reported plans to leave the profession within a year, and burnout rates reached 47.4% in Northern Michigan. Yet the statewide projected shortage remains modest at approximately 3% for RNs and 4% for LPNs, figures that likely mask more severe county-level imbalances given the geographic concentration documented above.

Bowser et al. (2025) further establish that market forces do not naturally correct this maldistribution. Examining nursing workforce growth across 376 metropolitan statistical areas from 2012 to 2022, they find no statistically significant relationship between RN or nurse practitioner growth and degree of medical underservice after controlling for income, employment, race, population size, and lagged wages. Growth instead tracked higher household income and stronger local labor markets, indicating that supply is being actively directed away from the highest-need areas. Notably, their analysis excludes nonmetropolitan settings, where, as Stratton et al. (1995) document, over 92% of federally designated nurse-shortage counties were concentrated, meaning the MSA-level null result likely understates the true degree of geographic maldistribution. Whether state-level figures similarly obscure finer county-level shortfalls within Michigan is precisely the question this paper addresses empirically in Section 6.

A second strand of literature asks why persistent vacancies continue despite decades of wage growth, a question that anticipates the unidentified wage channel discussed as a limitation in Section 7. Labor supply is characterized by modest responsiveness to wages at the market level (Askildsen et al., 2003) and near-inelastic responsiveness at the level of individual employers (Staiger et al., 2010). Early estimates focused on hours worked among currently employed nurses, with Askildsen et al. (2003) finding wage elasticities clustered around 0.2, suggesting that moderate wage increases produce only modest supply responses. Hanel et al. (2014) show that this framing understates the inelasticity problem by ignoring the extensive margin. Specifically, once the decision to enter or exit nursing from other occupations is incorporated, the unconditional wage elasticity of hours in nursing rises to 1.37, far above the intensive-margin estimate of 0.24. The policy implication is direct: wage increases are more effective at drawing nursing qualification holders back into the profession than at extracting additional hours from the existing workforce, a distinction that matters considerably for Michigan given its aging RN cohort.

At the firm level, even market-level elasticities overstate responsiveness. Staiger et al. (2010) exploit the Nurse Pay Act of 1990 as a natural experiment and find short-run firm-level supply elasticities of 0 to 0.2, consistent with substantial monopsony power. The mechanism is geographic differentiation: transportation costs give individual employers wage-setting power over local nursing labor supply, a friction that is likely more severe in rural Michigan counties where a single hospital system may be the dominant employer. Konetzka et al. (2018) add that RN and LPN supply respond asymmetrically to labor market conditions, with RN staffing falling and LPN staffing rising during recessions, a pattern that complicates policy design given that the two credentials are imperfect substitutes. This documented asymmetry foreshadows the central empirical result of the paper, the LPN-RN asymmetry in the training-pipeline coefficients reported in Section 6, and suggests that the two credentials should not be expected to respond identically to either labor-market shocks or training-capacity policy. Taken together, this strand of the literature points toward two conclusions that shape the empirical approach below. Wage-based interventions have limited aggregate reach, and the degree of supply inelasticity varies across space in a predictable way, with rural counties facing structurally worse conditions due to monopsonistic labor market structure.

A third strand of literature addresses the difficulty of measuring demand, since observed healthcare utilization is not equivalent to underlying healthcare demand. Utilization reflects not only medical need, but also whether care is available, affordable, geographically accessible, and acceptable to patients. Andersen (1995) provides the canonical framework for this distinction, decomposing healthcare use into predisposing characteristics, enabling factors, and need factors. For county-level nurse demand, this implies that demographic and morbidity measures may better capture structural need than realized utilization alone, a point that directly informs the demand proxies constructed in Section 4.

This distinction is especially important in rural and underserved areas, where low utilization may reflect constrained access rather than low need. Reilly (2021) emphasizes that rural healthcare access is shaped by availability, affordability, acceptability, accessibility, provider shortages, insurance coverage, poverty, and education. Evidence from a large Midwestern healthcare system is consistent with this concern. Nuako et al. (2022) find that patients using rural clinics had worse measured health status than patients using urban clinics, yet had lower outpatient utilization after adjusting for age, race, ethnicity, gender, smoking status, and health status. Utilization-based demand measures may therefore understate need precisely in the counties most likely to face access barriers.

The opposite concern arises in provider-rich areas. The supplier-induced demand literature argues that healthcare providers may influence utilization because patients rely on clinical expertise and face information asymmetries. Evans (1974) formalizes this concern, showing why healthcare demand models are incomplete when they ignore provider behavior. Yet empirical support for supplier-induced demand is limited. Grytten and Sørensen (2001), studying primary care physicians in Norway, find that neither salaried nor contract physicians increased output in response to greater physician density, suggesting mixed evidence of this behavior. Taken together, this strand of the literature suggests that realized utilization is a noisy proxy for RN and LPN demand. In underserved counties, utilization may be too low because care is difficult to obtain; in provider-rich counties, utilization may partly reflect supply-side conditions. To partially alleviate this issue, our analysis measures nursing demand primarily through demographic and morbidity-based indicators, including age structure, disability, poverty, and long-term-care need, as described further in Section 4.

A final strand of literature reinforces this paper's central pattern, that aggregate workforce projections can appear manageable while masking severe geographic mismatch. Early state-level forecasting work by Juraschek et al. (2012) used projected changes in population size and age to model RN supply and demand across the fifty states, finding that the number of states receiving a grade of "D" or "F" for RN shortage severity would rise from five in 2009 to thirty by 2030, with a projected national deficit approaching one million RN jobs. Zhang et al. (2018) updates this state-level forecasting approach and similarly emphasizes that demographic aging is a primary driver of projected demand. These studies provide the national benchmark for the present analysis, but their state-level scale leaves the county-level variation unresolved, which matters for Michigan policy.

More recent cohort-based projections refine the supply side of this problem. Auerbach et al. (2024) decomposes changes in the RN workforce into population, cohort, and age effects, distinguishing entry patterns across birth cohorts from retirement-related exits later in the life cycle. Using post-COVID data through 2022, they project the national RN workforce reaching approximately 4.56 million FTEs by 2035, close to pre-pandemic forecasts, with growth driven primarily by RNs aged 35 to 49. The implication is that aggregate recovery in the national RN workforce does not necessarily resolve local shortages. A growing national workforce may still fail to reach counties where aging, morbidity, and weak labor-market conditions produce the greatest need.

The HRSA's simulation-based projections make this distributional concern explicit. Nonmetro areas are projected to face RN shortages of 24 percent in 2028, 18 percent in 2033, and 11 percent in 2038, compared with 5, 4, and 2 percent respectively in metro areas over the same intervals (National Center for Health Workforce Analysis, 2025). Michigan is especially exposed to this pattern, with the HRSA projecting an 18 percent RN shortage in the state by 2038, ranking fourth nationally. Thus, the forecasting literature points to the same conclusion as the geographic and labor-supply literatures reviewed above. The relevant policy problem is not simply whether the nursing workforce is large enough in aggregate, but whether supply is located in the counties where demographic demand and structural disadvantage are rising fastest. The conceptual model and empirical framework that follow are designed to identify that mismatch within Michigan and to characterize the causal factors responsible for it.

3 Conceptual Model

Building on the literature reviewed above, we model county-level RN and LPN supply as the equilibrium outcome of a local healthcare labor market in which structural demand and structural supply conditions vary systematically with observable county characteristics. The framework is designed in particular to accommodate the empirical pattern documented in Section 6, where demographic demand pressure (the share of the population aged 75 and over) and local training pipelines operate on RN and LPN supply through distinct channels, and where one channel (local LPN training) is detectable within counties while the other (local RN training) is not. The idea is that persistent shortages reflect a geographic mismatch, in which counties with the greatest nursing need are systematically the least equipped to attract and retain nurses. Understanding the sources of that mismatch allows our results to inform targeted policy intervention.

On the demand side, counties differ in their structural need for nursing services as a function of population age structure and morbidity burden. We follow Andersen (1995) in distinguishing predisposing characteristics, enabling factors, and need factors as the components of healthcare demand, and proxy structural need using demographic and morbidity-based measures rather than realized utilization. This distinction is important because in rural counties where access barriers are pronounced, observed utilization systematically understates true need (Reilly, 2021), so a utilization-based demand measure would understate the severity of shortages in the counties facing the most acute access problems. Older, sicker, and more economically vulnerable populations generate higher structural demand for nursing care, independent of whether that demand is being met.

On the supply side, equilibrium nurse density in a county reflects both the location decisions of individual nurses and the wage-setting behavior of local employers. These two forces interact to produce a nurse supply that is not driven by patient need but by the structure of local labor and economic conditions. Nurses sort toward counties that offer higher wages, greater professional opportunity, and better amenities, which in practice means sorting toward urban and economically dynamic areas. This sorting is compounded by monopsonistic wage-setting in rural hospital markets: where a single employer or small number of employers dominate local hiring, equilibrium wages are suppressed below competitive levels, further discouraging nurse entry and retention (Staiger et al., 2010). Rural counties therefore face a compounding disadvantage on the supply side, both through the sorting decisions of nurses and through the structural inability of thin labor markets to sustain competitive wages. This sorting channel is also the lens through which Section 6 interprets the null relationship between local RN training output and local RN supply, since RNs are the more geographically mobile credential and are more able to act on these sorting incentives across county lines.

Structural disadvantage mediates both channels simultaneously. Counties with higher poverty rates, lower educational attainment, and weaker economic bases face elevated structural demand through higher chronic disease burden and disability rates, while facing depressed structural supply through lower wages and reduced capacity to attract workers. This dual pressure is the mechanism through which the geographic alignment between HPSAs and social disadvantage documented by Streeter et al. (2020) emerges at the county level in Michigan. It also implies that the nature of the shortage differs across counties: some face primarily a demand problem driven by a rapidly aging population, others face primarily a supply problem driven by economic distress and geographic isolation, and many face both. Distinguishing these cases is essential for policy, because they call for fundamentally different interventions.

This framework generates the testable predictions that organize the empirical framework in Section 5 and the results in Section 6. Counties with older populations and higher chronic disease prevalence should exhibit lower RN and LPN supply per capita, consistent with the mismatch hypothesis that counties with greater structural need are not attracting commensurate nurse supply and that local labor markets are not self-correcting for demographic demand pressure. Counties with higher poverty rates, greater rurality, and greater distance from nursing training programs should exhibit independently lower supply, reflecting structural barriers to nurse recruitment and retention. The empirical framework that follows tests these predictions using county-level panel data and translates the estimated coefficients into a county-by-county characterization of shortage type, providing a basis for geographically targeted policy recommendations.

4 Data

The analysis is built on a county-year panel covering all 83 Michigan counties over the period 2012 to 2023, excluding 2018 due to a gap in the nursing licensure data availability. The unit of observation is the county-year, yielding up to 913 observations for outcomes and controls with full coverage. All outcome variables are rate-normalized to county population, and the primary outcomes are registered nurses (RNs) per 10,000 residents and licensed practical nurses (LPNs) per 10,000 residents, with secondary denominators scaled to the resident population aged 65 and over where the analysis concerns elder care supply specifically. Two features of this panel construction recur in the discussion of limitations in Section 7: licensure is recorded at the nurse's county of residence rather than place of work, and several regressors share a common population denominator with the outcome variables. Summary statistics for the variables used in the regressions are reported in Table 1.

Table 1: Summary statistics, regression sample

Variable	N	Mean	SD	Min	P25	Median	P75	Max
RN per 10,000 residents	913	123.8	32.4	40.1	101.4	120.9	142.7	229.9
LPN per 10,000 residents	911	30.7	16.4	7.03	19.0	26.8	38.2	101.5
RN postings per 10,000	912	26.9	23.2	0.00	10.7	20.9	35.8	147.0
LPN postings per 10,000	912	5.35	5.48	0.00	2.15	3.94	6.93	59.3
Share aged 75+ (pp)	913	8.50	2.25	4.50	6.94	7.85	10.0	16.4
Share aged 65–74 (pp)	913	11.8	3.31	5.21	9.33	11.2	13.8	22.1
Population growth rate	913	0.00	0.01	−0.12	0.00	0.00	0.00	0.16
Hospital beds per 1,000	913	2.14	2.00	0.00	0.55	1.66	3.17	11.8
NH beds per 1,000 (65+)	913	0.01	0.13	0.00	0.00	0.00	0.00	2.37
Poverty rate (pp)	913	15.1	4.18	4.86	12.2	15.1	17.5	32.1
Bachelor's+ share (pp)	913	21.8	8.79	8.22	15.6	19.3	25.0	58.1
IPEDS RN grads/10k ($t-1$)	913	4.61	8.46	0.00	0.00	0.00	6.55	62.0
IPEDS LPN grads/10k ($t-1$)	913	1.58	4.83	0.00	0.00	0.00	0.22	53.0

Notes: County-year observations, 2012–2023 (excluding 2018). Outcomes are residence-based LARA licensure counts per 10,000 residents. Share and rate variables are in percentage points (pp). IPEDS completions are lagged one year. The outcome variables vary far more across counties than within counties over time, which motivates the within-county identification strategy in Section 5 and the between-county comparison reported in Section 6.

The primary source for nurse supply is the Michigan Department of Licensing and Regulatory Affairs (LARA), which maintains the statewide registry of active RN and LPN licenses. Licensure counts are recorded at the nurse's county of residence. The licensure data are supplemented with employer-reported occupational employment estimates from the U.S. Bureau of Labor Statistics Occupational Employment and Wage Statistics (BLS OEWS) program, providing annual county-level counts of employed RNs and LPNs alongside median and percentile wage figures. Because LARA licensure records active licenses (including nurses not currently employed) while OEWS records filled positions, the two series measure distinct margins of supply and are used separately in the analysis rather than combined. Wage measures are drawn exclusively from OEWS and enter only the supplementary wage specifications discussed in Section 7, where we report that the wage channel is not separately identified in the Michigan-only sample. OEWS does not publish county-level wage estimates for every county-year combination; suppressed cells are treated as missing, and the analysis accounts for this through the available sample rather than imputation. Posting-level wage data from Lightcast (formerly Burning Glass Technologies) are used as a supplementary measure of employer demand and are not used in the primary regression specifications.

Population structure and socioeconomic context are drawn from the American Community Survey (ACS) five-year estimates. Five-year pooled estimates are preferred over one-year estimates for county-level analysis because the one-year ACS does not release reliable small-area figures for counties with populations below 65,000, and the majority of Michigan's 83 counties fall below that threshold. The primary demographic demand proxy, motivated by the conceptual model in Section 3, is the share of county population aged 75 and over, chosen because this age group is the dominant driver of long-term and skilled nursing facility demand (see Juraschek et al., 2012). Additional controls include the poverty rate, the share of adults holding a bachelor's degree or higher, and the unemployment rate, all drawn from ACS five-year estimates at the county level.

Hospital bed capacity is constructed from the CMS Healthcare Cost Report Information System (HCRIS), which provides annual facility-level reports that are aggregated to the county level by summing licensed bed counts across all acute-care facilities within each county. Counties with no acute-care hospital in a given year are assigned a value of zero. Nursing home bed capacity is drawn from the CMS Nursing Home Provider Information file and is expressed as beds per 1,000 county residents aged 65 and over, consistent with the denominators used in the elder care literature. Health Professional Shortage Area (HPSA) designations and additional facility counts are drawn from the HRSA Area Health Resources File (AHRF), which aggregates federal administrative data at the county level and is updated annually.

The supply-side training pipeline, central to the empirical framework in Section 5 and to the headline LPN-RN asymmetry reported in Section 6, is measured using degree and certificate completions data from the National Center for Education Statistics (NCES) Integrated Postsecondary Education Data System (IPEDS) Completions Survey. Annual counts of nursing completions at the two-digit CIP code level (CIP 51.38 for registered nursing and CIP 51.39 for practical nursing) are aggregated to the county of the awarding institution and expressed as completions per 10,000 county residents. Because institutional output in year t enters the licensed workforce in subsequent years, this variable enters the regression lagged by one year. The training pipeline is the mechanism variable most central to the within-county identification strategy because it exhibits meaningfully more year-to-year variation than the demographic controls.

County-level chronic disease prevalence and population health measures are drawn from the County Health Rankings and Roadmaps program, which synthesizes administrative and survey data into annual county-level composites. The primary health status variable used in the descriptive analysis and typology construction is the share of county residents reporting fair or poor health, which proxies for structural nursing demand that goes beyond the elderly population share.

Rural classification follows the USDA Economic Research Service Rural-Urban Continuum Codes (RUCC), which assign each county a nine-point ordinal score based on population size and adjacency to metropolitan areas. Urban Influence Codes (UIC) are used as an alternative rurality measure in robustness checks. County-to-metropolitan-area linkages are established using the Office of Management and Budget (OMB) Core-Based Statistical Area (CBSA) delineation files. Because OMB periodically revises CBSA boundaries, the 2013 and 2023 delineation vintages are compared to ensure that the panel does not inadvertently capture reclassification as within-county change.

All sources are merged on a county Federal Information Processing Standards (FIPS) code and year key. The 2021 cross-section from the Michigan Public Health Institute (MPHI) nurse licensure survey provides additional licensure counts for that year and is merged as a supplementary column rather than replacing the LARA time series. Keweenaw County (FIPS 26083) has a suppressed LPN count in the MPHI cross-section and is assigned a missing value rather than zero for that cell. As Table 1 indicates, the primary source of variation in the outcome variables is cross-sectional rather than within-county over time, which motivates the attention paid to within-county identification in Section 5. The geographic distribution of nurse supply across Michigan counties is shown in Figures 1 and 2, the joint distribution of supply and demand outcomes in Figure 3, and the relationship between nurse supply and aging demand in Figure 4.

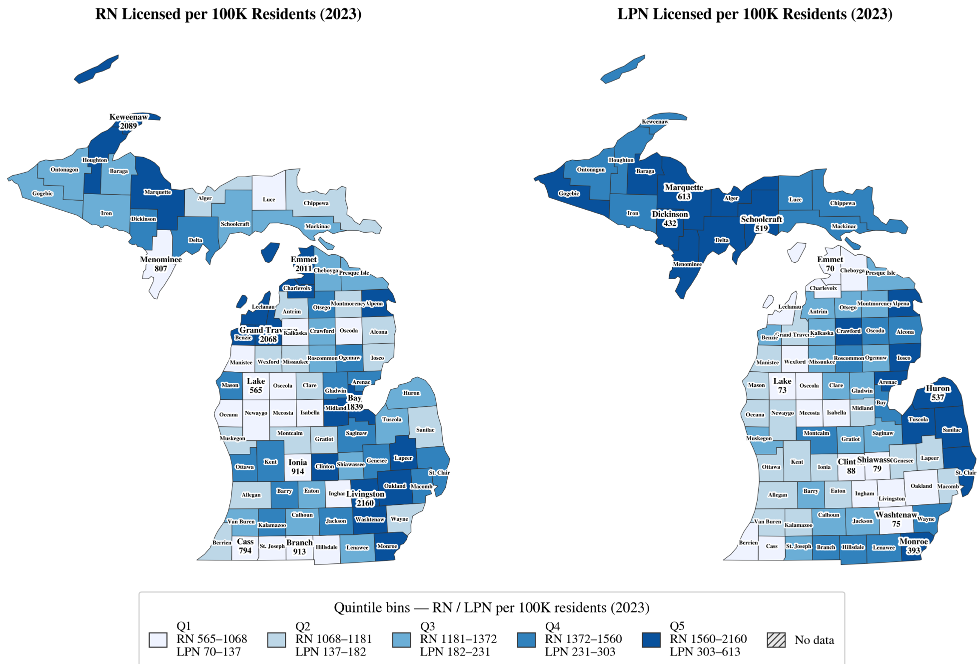

Figure 1: County-level supply of registered nurses and licensed practical nurses across Michigan, 2023, shown as population-scaled quintiles. Darker shading indicates counties in higher quintiles of nurse supply; lighter shading indicates lower relative supply. Source: LARA licensure records.

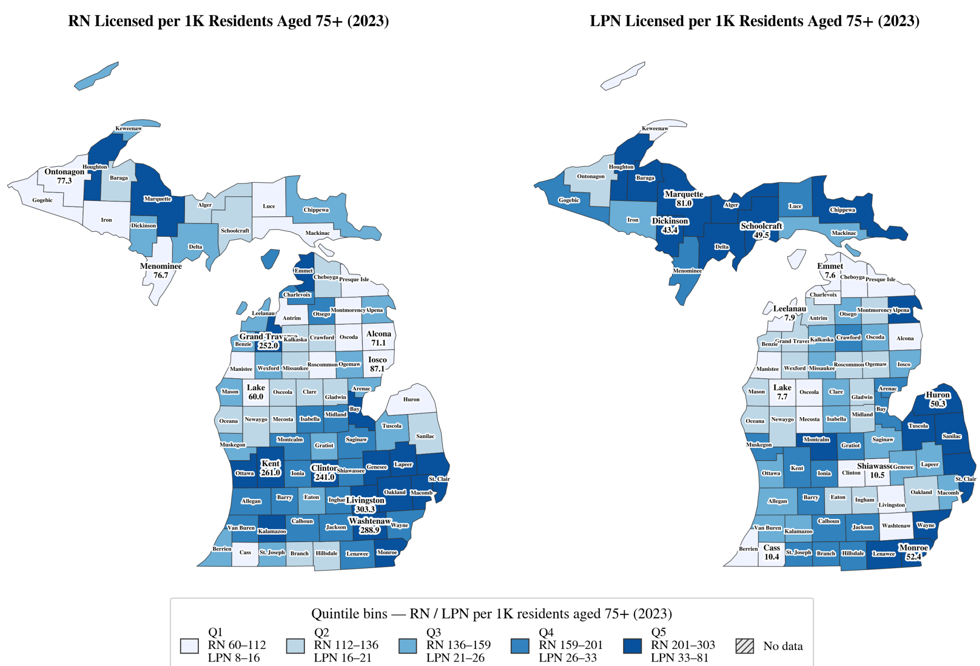

Figure 2: County-level RN and LPN supply scaled to the older-adult population: nurses per resident aged 75 and over across Michigan, 2023, shown as quintiles. Source: LARA licensure records; ACS five-year estimates.

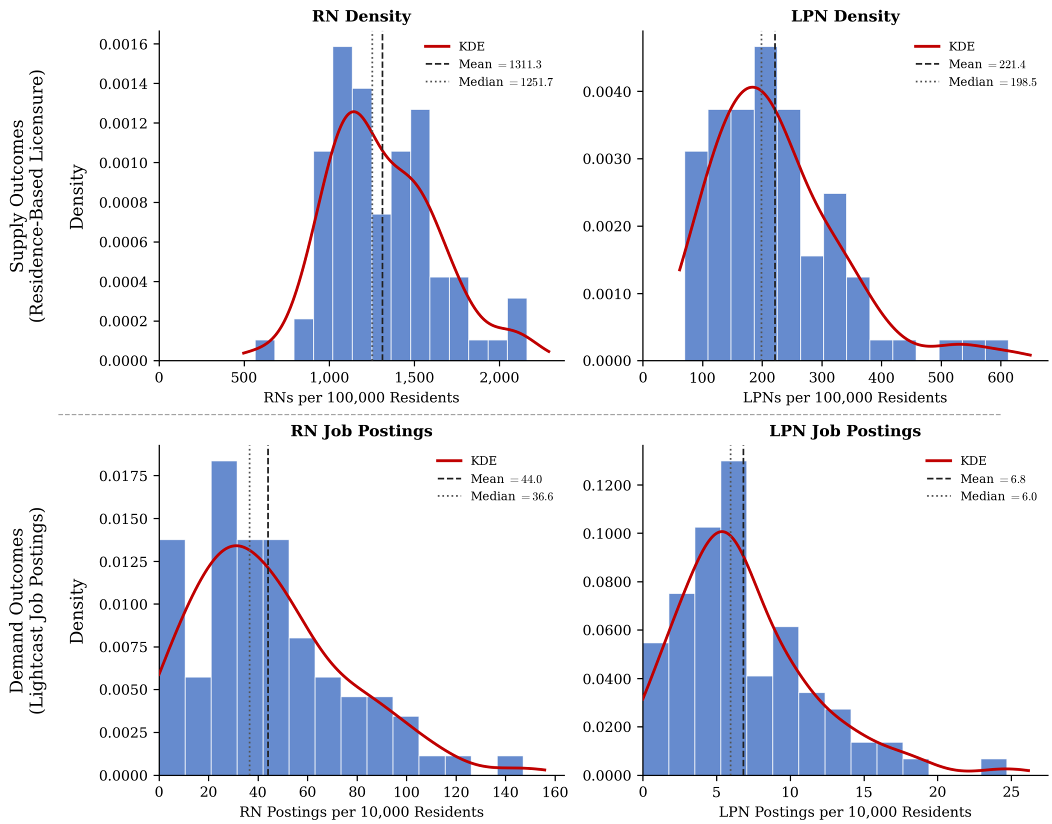

Figure 3: Distribution of nurse supply and employer demand outcomes, Michigan counties, 2023.

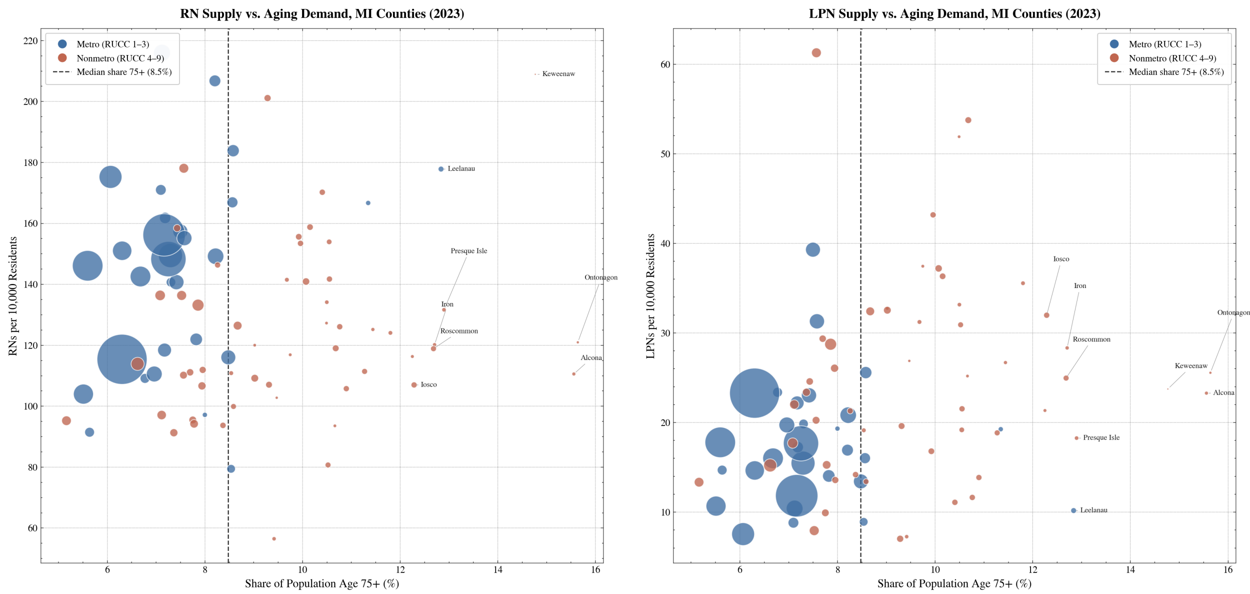

Figure 4: RN and LPN supply versus the share of population aged 75 and over across Michigan counties, 2023. Bubble size is proportional to county population, and color distinguishes metropolitan from nonmetropolitan counties. Counties in the lower-right region face the dual burden of high aging demand and low nurse supply. Source: LARA licensure records; ACS five-year estimates.

5 Empirical Framework

The empirical strategy estimates the conditional within-county relationship between county nurse supply and the structural demand, pipeline, and disadvantage measures identified in the conceptual model of Section 3. The core estimating equation is

$$Y_{it} \;=\; \alpha_i + \lambda_t + \beta_1 D_{it} + \beta_2 P_{i,t-1} + \beta_3 K_{it} + \beta_4 X_{it} + \beta_5 W_{it} + \varepsilon_{it}, \tag{1}$$

where Y_{it} is the number of registered nurses (RNs) or licensed practical nurses (LPNs) per 10,000 residents in county i in year t, measured from Michigan Department of Licensing and Regulatory Affairs (LARA) residence-based licensure files described in Section 4. The county fixed effect α_i absorbs all time-invariant unobserved county heterogeneity (geographic remoteness, persistent settlement patterns, historical institutional infrastructure). The year fixed effect λ_t absorbs Michigan-wide shocks common to all 83 counties (state policy changes, COVID-19 disruption to clinical placements, federal scholarship cycles). Standard errors are clustered at the county level.

D_{it} contains structural demand shifters: the share of the population aged 75 and over and the share aged 65 to 74, measured from American Community Survey five-year estimates, together with the population growth rate. The aging shares are entered separately because the oldest-old cohort is the dominant driver of long-term-care and inpatient nursing demand, whereas the younger-elderly cohort is more commonly served in lower-intensity ambulatory settings. Following Andersen (1995) and motivated by the access-barrier critique of utilization-based demand proxies in Reilly (2021), we deliberately use stock measures of population need rather than realized utilization, consistent with the conceptual model's demand-side channel.

$P_{i,t-1}$ is the local nurse-training pipeline, measured as IPEDS completions in the relevant occupation per 10,000 county residents, lagged one year. The one-year lag places the regressor in the information set before the current-year outcome and addresses the most mechanical form of simultaneity. Section 6 reports robustness to three- and five-year lags motivated by typical RN (BSN) and LPN training horizons.

K_{it} is a vector of healthcare-capacity conditioning variables: hospital beds per 1,000 residents and nursing-home beds per 1,000 residents aged 65 and over (AHRF), and the count of Health Professional Shortage Area primary-care designations in the county. These variables are co-determined with the nurse workforce and are included as conditioning information rather than as causal estimands; we present a five-year-lag robustness check to assess whether the contemporaneous coefficients are driven by simultaneity (Section 6).

X_{it} contains structural disadvantage controls (county poverty rate and the share of adults aged 25 and over with a bachelor's degree or higher), which mediate both the demand and supply channels described in the conceptual model.

W_{it} is a county nursing wage measure, either the PUMS county median for the corresponding occupation or the OEWS MSA-level wage, entered in supplementary specifications. Because wages and nurse supply are jointly determined in local labor markets, OLS estimates of β_5 are confounded; we report wage specifications as biased baselines and then attempt to identify the supply elasticity using a shift-share instrument that interacts national occupation-level wage growth with each county's initial nursing employment share, following Bartik (1991).

Because the right-hand-side variables differ systematically in their exposure to endogeneity, we organize the discussion of Equation (1) around an explicit credibility ladder. Coefficients on the aging shares (`share_75plus` and `share_65_74`) and on the lagged pipeline $P_{i,t-1}$ are interpreted as causal-leaning under the assumption that demographic structure and prior-year completions are predetermined relative to current-year supply shocks. Coefficients on `poverty_rate` and `bachelors_plus_share` are interpreted as robust conditional associations; these variables are slow-moving but are not exogenous to local labor-market conditions. Coefficients on capacity (K_{it}) and HPSA designation are interpreted as conditioning information only: capacity is co-determined with workforce, and HPSA is mechanically tied to the shortage outcome itself. The Magnet-hospital indicator is excluded from all reported specifications as a bad control, since Magnet status plausibly reflects prior workforce strength rather than driving supply. Coefficients on wages are not separately identified in this design, as documented in Section 7.

The two-way fixed-effects specification is preferred over a cross-sectional approach for two reasons. First, it eliminates bias from time-invariant unobservables that are plausibly correlated with both nurse supply and the demand and disadvantage measures of interest. Second, the within-county identification it provides is more credibly tied to the policy question, namely what happens when a county's pipeline, demographics, or capacity change, than is cross-sectional comparison. To benchmark the sensitivity of our findings to this choice, we also estimate pooled OLS specifications on the same sample without county or year fixed effects, adding a constant and the county foreign-born population share, a time-invariant control that the county fixed effect would otherwise absorb. These pooled estimates recover the between-county variation that the fixed-effects design removes. Because dropping the county fixed effect re-exposes the model to the cross-sectional omitted-variable bias that motivated the two-way design, the pooled estimates are reported alongside the TWFE results in Section 6 as descriptive between-county benchmarks rather than as more credible estimates.

The shift-share instrument used to identify the wage channel is constructed as $Z_{it} = \sum_k s_{ik,2014} \cdot g_{kt}^{\text{nat}}$, where $s_{ik,2014}$ is county i's 2014 share of employment in industry k and g_{kt}^{nat} is the national wage growth rate in industry k. Identification requires both relevance (that initial industry composition transmits national wage shocks into local nursing wages) and exclusion (that initial industry shares affect nurse supply only through wages, conditional on the controls). As Section 6 documents, the first-stage F statistic is well below the conventional weak-instrument threshold in the Michigan-only sample; the wage coefficient is therefore not identified in this design. We retain the 2SLS specification for diagnostic transparency rather than for inference, and discuss the shift-share validity conditions of Borusyak et al. (2022); Goldsmith-Pinkham et al. (2020) as part of the limitations in Section 7.

6 Results

We organize the results around the three mechanism analyses that are identified in this design: baseline geographic distribution, the local training pipeline, and vacancy-pressure targeting. The mechanisms that are not separately identified, namely retention dynamics, employer financial capacity, and the spatial wage channel, are deferred to Section 7. All two-way fixed-effects (TWFE) specifications are estimated on the 83-county Michigan panel for 2012 to 2023 (excluding 2018) with county and year fixed effects and standard errors clustered at the county level. Because the demographic and socioeconomic controls vary far more across counties than within them, we report a pooled between-county OLS specification alongside the fixed-effects estimates. The bulk of within-county variance in nurse supply is not captured by the included regressors, so null within-county coefficients should not be read as evidence of zero effects.

6.1 Baseline Distribution Models

Table 2 reports the preferred baseline TWFE specification. The aging controls separate the population aged 75 and over, which generates the bulk of long-term-care and inpatient nursing demand, from the population aged 65 to 74, the younger-elderly cohort more commonly served in ambulatory settings. Additional controls include population growth, hospital and nursing-home capacity, poverty, educational attainment, and the one-year-lagged training pipeline.

Within-county increases in the 75-and-over share are positively and significantly associated with RN supply ($\hat{\beta} = 5.77$, $p = 0.002$). The 65-to-74 cohort shows no significant relationship with RN supply ($\hat{\beta} = -1.18$, $p = 0.566$), consistent with the oldest-old cohort being the dominant driver of acute-inpatient and long-term-care demand. For LPNs, neither aging cohort is significant within county, and the borderline negative coefficient on the 65-to-74 share ($\hat{\beta} = -1.88$, $p = 0.052$) is too imprecise to interpret structurally. As shown below, the LPN aging signal is recovered between counties and is simply not identified off within-county variation. The share of adults with a bachelor's degree or higher is positively and significantly associated with LPN supply ($\hat{\beta} = 0.79$, $p = 0.013$) and positive but imprecise for RN supply ($\hat{\beta} = 0.89$, $p = 0.185$), consistent with a labor-pool channel in which larger college-educated populations sustain a broader candidate pool for nursing licensure. Poverty shows no significant within-county association with either occupation.

Hospital and nursing-home bed capacity enter as conditioning variables rather than causal regressors. For LPN supply, both capacity measures are negative and highly significant ($\hat{\beta} = -2.54$ and -3.47, both $p < 0.001$), a pattern consistent with reverse causation, in which capacity is built where a nursing workforce already exists, rather than with a structural effect of capacity on supply. A five-year-lagged capacity specification supports this reading, since the hospital-bed coefficient for LPN supply collapses toward zero when lagged.

Between-county robustness. Because the aging, poverty, and education shares vary far more across counties than within them, with only 5 to 19 percent of their variance occurring within-county, the TWFE design is underpowered to detect their effects. Table 3 therefore

Table 2: Baseline distribution models (two-way fixed effects)

	RN per 10,000		LPN per 10,000	
Variable	Coef.	(SE)	Coef.	(SE)
Share aged 75+ (pp)	5.77**	(1.83)	1.02	(1.30)
Share aged 65–74 (pp)	−1.18	(2.06)	−1.88†	(0.97)
Population growth rate	−7.81	(38.47)	−49.13***	(13.55)
Hospital beds per 1,000	−1.09	(0.84)	−2.54***	(0.44)
NH beds per 1,000 (65+)	0.45	(1.04)	−3.47***	(0.52)
Poverty rate (pp)	0.69	(0.49)	0.07	(0.26)
Bachelor's+ share (pp)	0.89	(0.67)	0.79*	(0.32)
IPEDS RN grads/10k ($t-1$)	0.024	(0.119)	—	
IPEDS LPN grads/10k ($t-1$)	—		0.416***	(0.055)
County & year FE	Yes		Yes	
Clusters (county)	83		83	
Observations	913		911	
Within R^2	0.088		0.164	

reports a pooled between-county OLS column alongside the TWFE estimates. Several covariates that are insignificant within county are strongly significant between counties with the expected sign: the 75-and-over share for LPN supply ($\hat{\beta} = 7.14$, $p < 0.001$, versus 1.02 within) and the bachelor's-degree share for RN supply ($\hat{\beta} = 2.30$, $p < 0.001$, versus 0.89 within). This indicates the within-county nulls reflect limited identifying variation rather than the absence of a relationship. We treat the pooled estimates as descriptive between-county benchmarks subject to cross-sectional omitted-variable bias, the very bias the fixed-effects design removes, and not as more credible than the within-county estimates. Two patterns reinforce this reading. First, the capacity coefficients reverse sign between the within and between estimators, consistent with bed-rich counties being larger health-care markets cross-sectionally even though within-county capacity growth does not raise supply. Second, the LPN training-pipeline coefficient is positive and significant in both the within (0.42) and between (0.67) specifications, so the headline result is robust to identification strategy.

6.2 Training-Capacity Models: The Headline Result

The training-capacity models test whether within-county variation in lagged IPEDS nursing completions predicts current nurse supply. This is the paper's headline causal-leaning result, because lagged completions are predetermined relative to the current-year outcome. Table 4 reports the pipeline coefficient across specifications.

Table 3: Within-county (TWFE) vs. between-county (pooled OLS) baseline estimates

	RN per 10,000		LPN per 10,000	
Variable	TWFE	Pooled	TWFE	Pooled
Share aged 75+ (pp)	5.77**	4.06*	1.02	7.14***
Share aged 65–74 (pp)	−1.18	−2.12	−1.88†	−4.48***
Hospital beds per 1,000	−1.09	3.18*	−2.54***	0.93†
NH beds per 1,000 (65+)	0.45	−5.82	−3.47***	8.24***
Poverty rate (pp)	0.69	−2.23***	0.07	−0.12
Bachelor's+ share (pp)	0.89	2.30***	0.79*	−0.46*
IPEDS RN grads/10k ($t-1$)	0.024	0.239	—	—
IPEDS LPN grads/10k ($t-1$)	—	—	0.416***	0.665***
County & year FE	Yes	No	Yes	No

Notes: Pooled columns add a constant and recover between-county variation; they are descriptive benchmarks subject to cross-sectional omitted-variable bias, not more credible than the TWFE estimates.

In the baseline specification without a lagged dependent variable, an additional LPN graduate per 10,000 county residents in the prior year is associated with 0.416 additional LPNs per 10,000 current-year residents ($SE = 0.055$; asymptotic clustered $p < 0.001$). With $G = 83$ clusters, asymptotic standard errors are at the lower bound of reliability, so we report the wild-cluster bootstrap (Rademacher weights, $B = 999$, null imposed), which gives a p-value of 0.057. This is borderline significant at 5 percent and clearly significant at 10 percent. The honest characterization is that the LPN training-to-supply link is positive and precisely estimated under asymptotic inference but more marginal under finite-cluster bootstrap inference appropriate for $G = 83$. The coefficient is robust across specifications: adding a lagged dependent variable attenuates it to 0.264 ($p < 0.001$), as expected from Nickell bias with $T \approx 8$; it survives a two-year lag (0.259, $p = 0.004$); and a log-count specification with log(population) as an offset, which addresses correlated measurement error when both outcome and regressor share a population denominator, yields a log-log elasticity of 0.033 ($p = 0.006$).

The analogous RN pipeline coefficient is near zero in every specification: 0.024 ($p = 0.843$) in the baseline, 0.009 ($p = 0.816$) with a lagged dependent variable, and null across all urbanicity subsamples. This RN-LPN asymmetry is the central empirical finding. We interpret it as reflecting greater geographic mobility among RN graduates, whose longer training horizon and more transferable credential lead them to disperse across county lines, so that residence-based licensure at the county level cannot detect a training-to-supply link. LPN graduates, trained primarily through 12-month community-college certificate programs, exhibit substantially higher local-retention rates consistent with the 0.416 coefficient.

Table 4: Training-pipeline coefficient across specifications

Specification	IPEDS LPN/10k ($t-1$) Coef.	(SE)	IPEDS RN/10k ($t-1$) Coef.	(SE)
Baseline (no lagged DV)	0.416***	(0.055)	0.024	(0.119)
Training spec (lagged DV)	0.264***	(0.079)	0.009	(0.040)
Two-year lag (baseline)	0.259**	(0.090)	—	
Log-count (common denom.)	0.033**	(0.012)	—	
Wild-cluster bootstrap p	0.057			—
Urbanicity (LPN pipeline)				
MSA only (51)	0.457**	(0.167)		
Rural only (32)[a]	0.181†	(0.095)		
Rural × pipeline	−0.124	(0.120)		

Notes: County and year FE, county-clustered SEs. Baseline uses the Table 2 controls; the training spec adds a lagged dependent variable, lagged entry-age share, and unemployment. Log-count uses $\log(1 + \text{LPN count})$ with log(pop) offset (a log–log elasticity). [a] identified off a single rural program county. †$p < .10$, *$p < .05$, **$p < .01$, ***$p < .001$.

Pre-trends and anticipation. A leads-and-lags specification (Table 5) finds that one- and two-year leads of LPN completions are near zero and insignificant (lead 1: −0.043, $p = 0.688$; lead 2: +0.158, $p = 0.195$), while both lags are positive and significant (lag 1: 0.204, $p = 0.010$; lag 2: 0.141, $p = 0.039$). The absence of pre-trends is inconsistent with anticipatory program expansion in response to observed local shortages and supports a forward-looking pipeline interpretation. A reverse-Granger regression of current completions on lagged LPN supply finds only borderline evidence of anticipatory seat adjustment (coefficient on supply two years prior: −0.016, $p = 0.024$), which we note as a caveat but do not treat as invalidating the result. The LPN pipeline coefficient is directionally consistent across the MSA (0.457, $p = 0.007$) and rural (0.181, $p = 0.059$) subsamples, but the rural estimate is identified off a single program county, since only 1 of 32 rural counties hosts an LPN program. A rural-interaction test cannot reject homogeneity ($p = 0.305$) but is severely underpowered, so expanding LPN training to currently unserved rural counties is a policy extrapolation beyond the regressor's support.

6.3 Vacancy-Pressure Targeting

The vacancy-pressure models ask whether within-county changes in structural conditions predict active employer job postings (Lightcast). These are diagnostic targeting tools rather than causal models, because postings measure observable recruitment demand, under-represent small rural employers who recruit informally, and conflate nominal vacancies with effective shortages. They explain little within-county variation (within $R^2 = 0.043$ for RN

Table 5: Leads-and-lags (pre-trends) test, LPN pipeline

Term	Coef.	(SE)
IPEDS LPN ($t+2$)	0.158	(0.122)
IPEDS LPN ($t+1$)	−0.043	(0.106)
IPEDS LPN ($t-1$)	0.204**	(0.079)
IPEDS LPN ($t-2$)	0.141*	(0.068)

Notes: Two-way FE, county-clustered SEs. Insignificant leads ($t+1$, $t+2$) indicate no pre-trends; significant lags support a forward-looking pipeline. $^{*}p < .05$, $^{**}p < .01$.

postings and 0.016 for LPN postings), indicating that the structural determinants of posting behavior are largely absorbed by the county and year fixed effects. Within county, higher poverty ($\hat{\beta} = 1.54$, $p = 0.024$) and nursing-home bed density ($\hat{\beta} = 4.49$, $p < 0.001$) predict more RN postings, while lagged RN supply is negative but insignificant ($\hat{\beta} = -0.105$, $p = 0.167$) and no predictor is significant for LPN postings (Table 6). We therefore use postings not as evidence of a mechanism but as a descriptive signal for identifying counties with observable recruitment pressure, which are candidates for loan-repayment or placement programs.

Table 6: Vacancy-pressure models (postings per 10,000, TWFE)

	RN postings		LPN postings	
Variable	Coef.	(SE)	Coef.	(SE)
Share aged 75+ (pp)	−5.35*	(2.64)	−1.02	(0.83)
Share aged 65–74 (pp)	−2.02	(2.51)	−0.70	(0.64)
Population growth rate	52.47	(149.33)	−3.20	(22.89)
HPSA (primary care)	−9.94	(9.94)	0.75	(1.36)
Hospital beds per 1,000	1.61	(1.10)	0.10	(0.21)
NH beds per 1,000 (65+)	4.49***	(1.24)	−0.51	(0.34)
Poverty rate (pp)	1.54*	(0.68)	0.25	(0.26)
Bachelor's+ share (pp)	0.83	(0.55)	0.26	(0.23)
RN supply ($t-1$)	−0.10	(0.08)	—	
LPN supply ($t-1$)	—		−0.03	(0.06)
IPEDS RN grads/10k ($t-1$)	−0.021	(0.105)	—	
IPEDS LPN grads/10k ($t-1$)	—		0.022	(0.068)
County & year FE	Yes		Yes	
Clusters (county)	83		83	
Observations	828		826	
Within R^2	0.043		0.016	

6.4 Headline Interpretation

The principal finding is that local LPN training output, measured by lagged IPEDS completions, predicts subsequent within-county LPN supply at a magnitude consistent with local-retention rates of 40 to 60 percent. This result is robust across the inclusion of a lagged dependent variable, a two-year lag, a log-count alternative outcome, the pooled between-county specification, and the wild-cluster bootstrap ($p = 0.057$, $G = 83$). The analogous RN pipeline coefficient is null at every tested specification and urbanicity stratum. This asymmetry implies that LPN and RN shortages call for different instruments at different scales. Expanding community-college LPN capacity in shortage counties should generate measurable, locally retained supply gains, whereas RN maldistribution, driven by a more mobile workforce, requires training and retention policy coordinated at the regional or state level. The vacancy-pressure models add a county-targeting tool but no additional identified channel.

7 Limitations

Several limitations qualify the framework, spanning measurement, identification, statistical inference, and the set of mechanisms that this design can and cannot separate.

7.1 Measurement and Data Quality

The LARA licensure outcome measures nurses where they *reside*, not where they *work*. Nurses who live in one county and commute to a hospital in another are assigned to the residence county. Because nurse labor markets are regional, particularly in southeast Michigan, this introduces measurement error that does not average out across counties and that the county fixed effects cannot absorb. Workplace-based supply, the policy-relevant concept for staffing, is not directly identified by Equation (1), and this is our preferred interpretation of the RN training null. Relatedly, licensure counts include retired, inactive, and out-of-state-employed but Michigan-licensed nurses; without a full-time-equivalent adjustment, the per-capita measure conflates participation and effective hours. A further concern is that both the outcome and the IPEDS pipeline regressor are normalized by county population, so intercensal population revisions, especially in small rural counties, propagate as correlated measurement error. The log-count specification with a log(pop) offset (Table 4) addresses this for the preferred LPN pipeline coefficient.

7.2 Identification

The lag-one IPEDS variable is predetermined but is not exogenous if colleges expand seats in anticipation of locally visible shortages, since a county-specific demand shock persisting from $t-2$ into t would correlate with both prior-year completions and current outcomes. We test for this directly: the leads-and-lags specification shows no significant pre-trends (Table 5), and a reverse-Granger regression yields only a small, borderline coefficient on lagged supply (-0.016, $p = 0.024$). These tests weigh against, but do not decisively rule out, anticipatory expansion. The HPSA shortage-area indicator moves infrequently within counties and is partly endogenous to the outcome; under treatment-effect heterogeneity, the TWFE coefficient on a slowly-changing binary regressor is a weighted average of pairwise difference-in-differences comparisons with potentially negative weights (Goodman-Bacon, 2021; de Chaisemartin and D'Haultfœuille, 2020), so we report it descriptively only. The Magnet-hospital indicator was excluded from all reported specifications as a bad control, since Magnet status plausibly reflects prior workforce strength rather than driving supply. Finally, county-level estimation assumes independence across counties (SUTVA), yet cross-county commuting and training flows violate this assumption. Clustering at the county level corrects the variance but not the bias from spillovers, and a commuting-zone analysis would help bound this.

7.3 Inference and Power

The panel contains 83 counties. Cluster-robust standard errors are asymptotically valid as the number of clusters grows; with $N_g = 83$, finite-sample bias is plausible, so we report the wild-cluster bootstrap of Cameron et al. (2008) for the headline coefficient. Within-county R^2 values are modest (0.088 in the preferred RN specification and 0.164 in the LPN specification), reflecting that the bulk of within-county variation in licensure density is not captured by the regressors; the pooled between-county estimates (Table 3) partly address this by recovering the cross-sectional gradients the within design cannot detect. The IPEDS pipeline is identified off the minority of counties that host nursing programs, roughly 18 with an LPN program and 37 with an RN program of 83, so the estimated effect applies to the program-county subsample and does not directly support projections for program-free counties. The Michigan-only, 2012 to 2023 scope further limits external validity. The LPN training-to-supply coefficient is the single pre-specified focal hypothesis, while the remaining estimates are exploratory; we therefore do not apply an ad hoc multiplicity correction to the focal test, but caution that, given the number of specifications estimated, some borderline-significant exploratory results are expected by chance. Under the finite-cluster bootstrap the focal coefficient is itself only marginally significant ($p = 0.057$), which we report as the headline inference.

7.4 Mechanisms Not Separately Identified

Three of the five policy levers we examined did not yield an identified effect in this design, and we treat them as directions for future multi-state work rather than as findings. Workforce age-composition variables are available only for 2012 to 2017 and 2019 to 2020, excluding the 2021 pandemic-exit year, the period with the most structurally unusual retention dynamics. In the available window, no age-composition variable is significant for RN supply, and only the lagged LPN under-35 entry share is marginally positive ($\hat{\beta} = 0.096$, $p = 0.039$). Burnout, intent-to-leave, and part-time margins are not measured at the county-year level, and a supplementary part-time-share test was not estimable given only three post-2018 years of coverage; county-level burnout or staffing-intensity data would be required to identify a retention channel. Hospital operating margins (CMS HCRIS, 73 counties) show negative but insignificant associations with supply ($\hat{\beta} = -0.38$ for RN and -0.27 for LPN), and neither survives outlier or thin-year diagnostics, since the coefficients flip sign when the two thinly covered panel years or 14 extreme-margin pandemic-era observations are excluded; cleaner financial-health measures such as days cash on hand or bond ratings are needed. The shift-share instrument for local wages is weak in the Michigan-only sample (first-stage $F < 1$ in both occupations) and requires share- or shock-orthogonality conditions (Borusyak et al., 2022; Goldsmith-Pinkham et al., 2020) that industries employing nurse-occupation substitutes plausibly violate, so we treat the wage channel as unidentified rather than weakly identified. A nurse-practitioner substitution interaction is statistically present ($\hat{\beta} = -0.015$, $p = 0.004$) but, because NP density is co-determined with the local health system, is a conditional association rather than a substitution elasticity. A multi-state panel with stronger identifying variation is the natural path to recovering these channels.

8 Conclusion

This paper set out to determine whether Michigan's nursing workforce problem is better understood as an aggregate shortage or as a geographic mismatch, and the evidence points toward the latter. Across the mechanisms we examined, the clearest and most policy-relevant signal is the asymmetry between the RN and LPN training pipelines. Local LPN program output predicts subsequent within-county LPN supply at a magnitude consistent with local-retention rates of 40 to 60 percent, a result that survives a lagged dependent variable, a two-year lag, a log-count specification, a between-county pooled benchmark, and a finite-cluster wild-cluster bootstrap ($p = 0.057$). The analogous RN coefficient is null everywhere it is tested, consistent with a more geographically mobile RN workforce that residence-based licensure cannot tie back to county of training. The between-county estimates corroborate the aging-demand and education gradients that the within-county design is underpowered to detect, while the sign reversal of the capacity coefficients across the within and between estimators reinforces that bed capacity is co-determined with supply rather than causal.

This asymmetry implies that LPN and RN shortages call for different instruments at different geographic scales. Expanding community-college LPN capacity in shortage counties is a lever with direct, locally retained payoff, whereas RN shortages are unlikely to be resolved by county-level training expansion alone and will require training and retention policy coordinated at the regional or state level. The vacancy-pressure models, while not identifying a causal channel, remain useful as a targeting tool for counties where loan-repayment or placement programs are likely to have the greatest near-term impact. The retention, employer-capacity, and wage mechanisms were not separately identified in this design, owing to incomplete age-composition coverage, pandemic-distorted hospital accounting, and a weak Michigan-only wage instrument, and they are the priorities for a multi-state extension. More broadly, the modest within-county R^2 values across all specifications are themselves informative, since the bulk of the geographic variation in Michigan's nurse supply is cross-sectional rather than dynamic, reflecting deep and slow-moving differences across counties rather than year-to-year fluctuations that policy can readily move. Closing Michigan's geographic mismatch in nursing care is therefore a long-horizon undertaking, and the local LPN training pipeline identified here is one of the few levers with a demonstrated within-county effect.

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